EXPERIMENT – 9

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Design input forms that validate data (email, phone number) and display error messages using HTML, CSS, and JavaScript.

AIM:

To design and implement an input form that validates user data such as email and phone number using HTML, CSS, and JavaScript with Validator.js, and to display appropriate error messages along with visual feedback.

Step 1: Creating the HTML Form

Description:
An HTML form is created with input fields for email and phone number. Error message placeholders are added below each field to display validation messages dynamically.

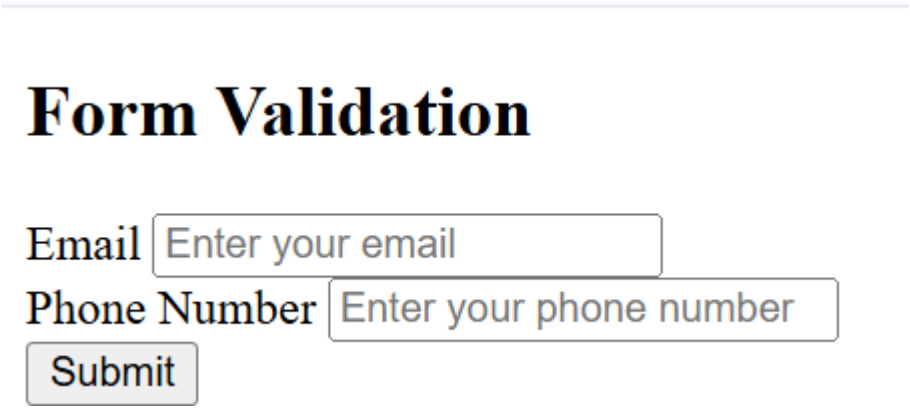
Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Form Validation - Experiment 9</title>
   <!--<link rel="stylesheet" href="style.css">-->
</head>
<body>
```

```
    <div class="container">
        <h2>Form Validation</h2>

        <form id="myForm">
            <div class="form-group">
                <label>Email</label>
                <input type="text" id="email" placeholder="Enter your email">
                <small class="error" id="emailError"></small>
            </div>

            <div class="form-group">
                <label>Phone Number</label>
                <input type="text" id="phone" placeholder="Enter your phone
number">
                <small class="error" id="phoneError"></small>
            </div>

            <button type="submit">Submit</button>

        </form>
    </div>
    <script
src="https://cdnjs.cloudflare.com/ajax/libs/validator/13.9.0/validator.min.js">
/script>
    <script src="validator.js"></script>

</body>
```

```
</html>
```

Output:



Step 2: Styling using CSS

Description:
CSS is applied to enhance the visual appearance of the form. A modern card layout is used with gradients, rounded corners, and smooth animations. Input focus effects and button hover effects improve user interaction.

Key Enhancements:

- Gradient background
- Centered card layout
- Smooth animation (fade-in)
- Hover effects on button
- Input focus glow

Code:

```
* {
```

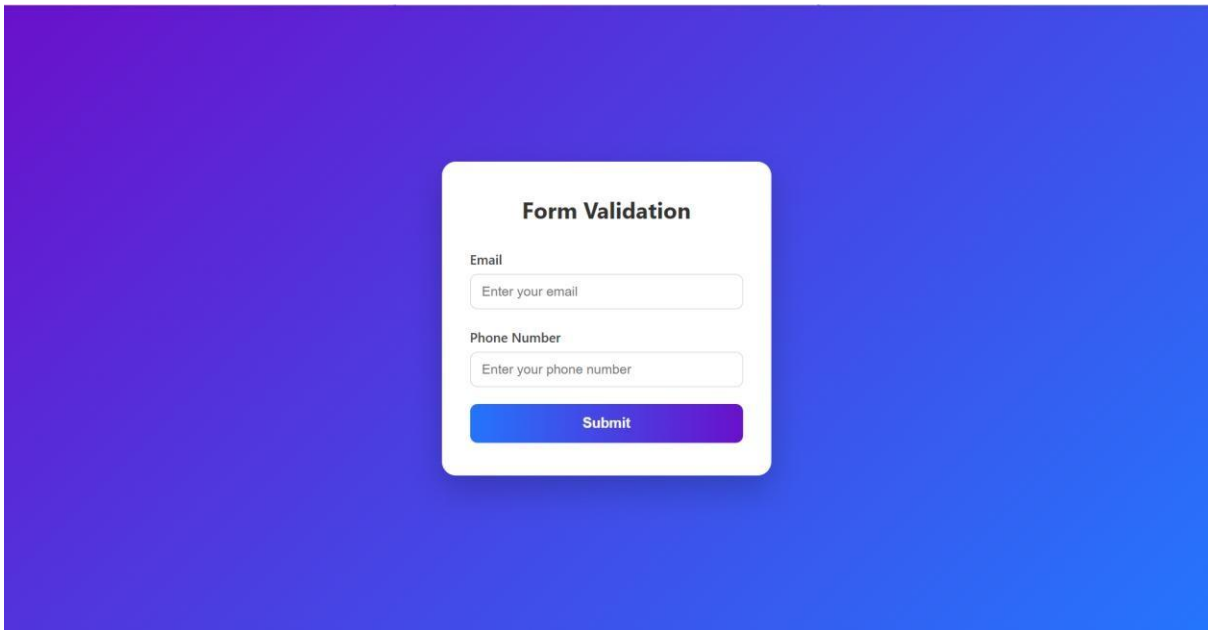
```
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    background: linear-gradient(135deg, #6a11cb, #2575fc);
    height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
}
.container {
    background: #ffffff;
    padding: 35px 30px;
    border-radius: 15px;
    width: 350px;
    box-shadow: 0 15px 35px rgba(0, 0, 0, 0.2);
    animation: fadeIn 0.8s ease-in-out;
}

h2 {
    text-align: center;
    margin-bottom: 25px;
    color: #333;
```

```
}
.form-group {
    margin-bottom: 18px;
}
label {
    font-weight: 600;
    font-size: 14px;
    color: #444;
}
input {
    width: 100%;
    padding: 10px 12px;
    margin-top: 6px;
    border-radius: 8px;
    border: 1px solid #ccc;
    transition: 0.3s ease;
    font-size: 14px;
}

input:focus {
    outline: none;
    border-color: #2575fc;
    box-shadow: 0 0 6px rgba(37, 117, 252, 0.4);
}
button {
    width: 100%;
```

```
    padding: 12px;
    background: linear-gradient(to right, #2575fc, #6a11cb);
    color: white;
    border: none;
    border-radius: 8px;
    font-size: 15px;
    font-weight: bold;
    cursor: pointer;
    transition: 0.3s ease;
}

button:hover {
    transform: translateY(-2px);
    box-shadow: 0 8px 20px rgba(0,0,0,0.2);
}
.error {
    color:    #e74c3c;
    font-size:    12px;
    margin-top: 4px;
}
input.success {
    border-color: #2ecc71;
}
input.error-border {
    border-color: #e74c3c;
}
```

```
@keyframes fadeIn {
  from {
    opacity: 0;
    transform: translateY(20px);
  }
  to {
    opacity: 1;
    transform: translateY(0);
  }
}
```

Output:



Step 3: Adding JavaScript Validation

Description:
JavaScript along with Validator.js is used to validate the email and phone number fields. Error messages are displayed dynamically for invalid inputs.

Additionally, visual feedback is provided:

- Red border for invalid inputs
- Green border for valid inputs

This improves user experience by clearly indicating input status.

Validation Used:

- validator.isEmail() for email
- validator.isMobilePhone() for phone

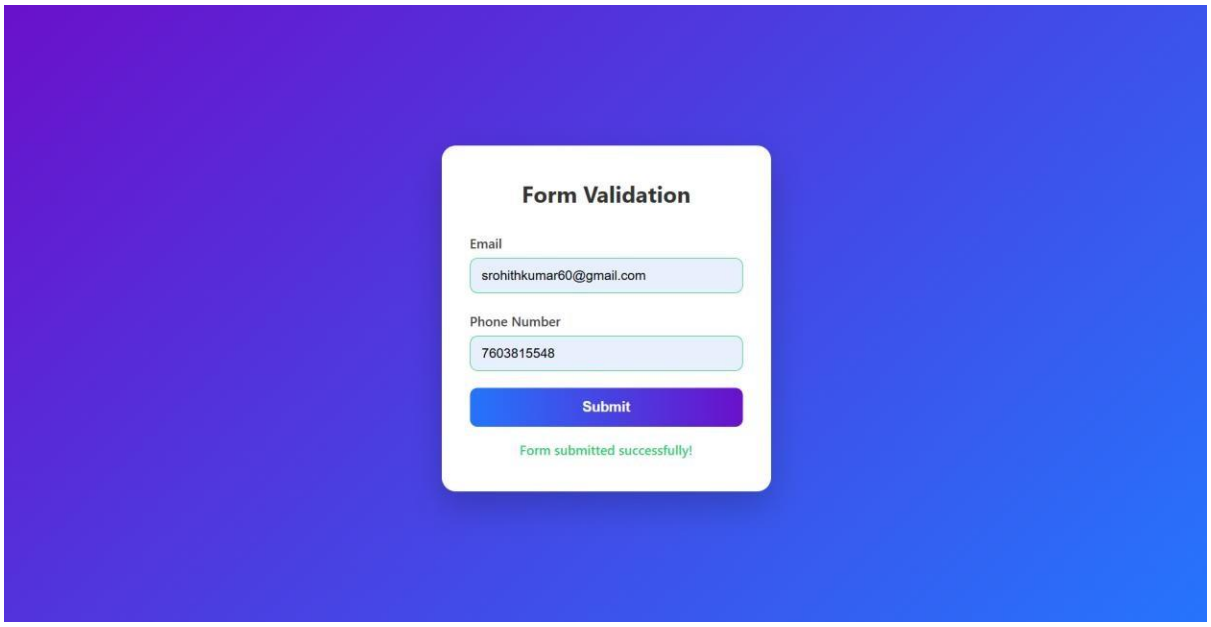
Code:

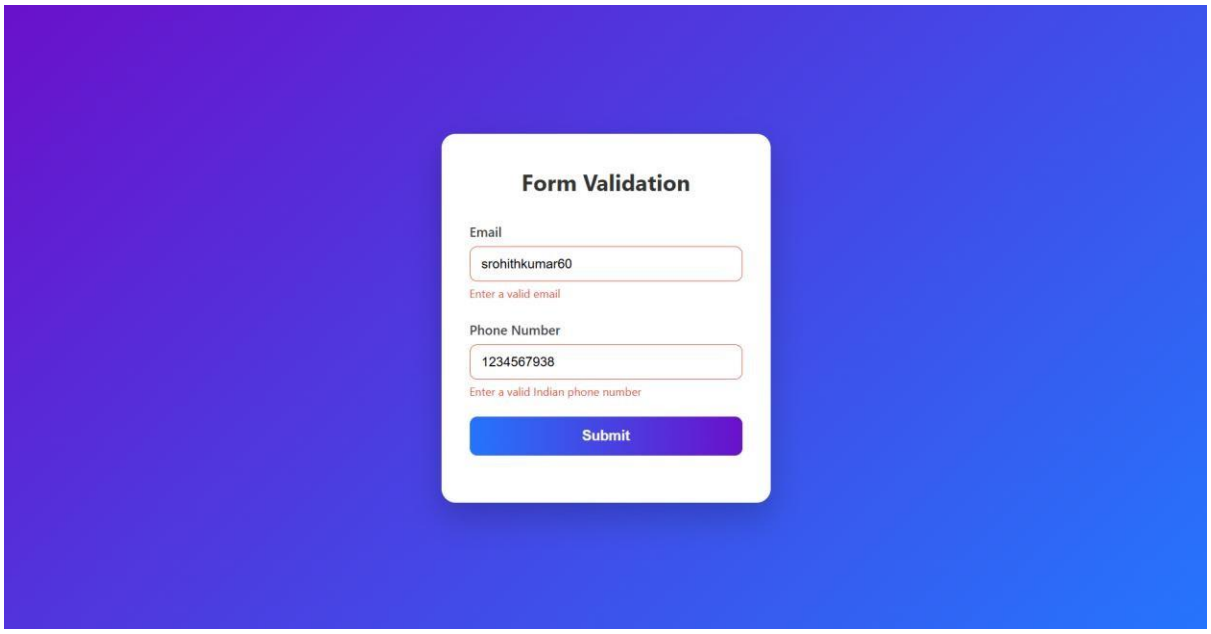
```
const form = document.getElementById("myForm");


form.addEventListener("submit", function(e) {
  e.preventDefault();


  let email = document.getElementById("email").value.trim();
  let phone = document.getElementById("phone").value.trim();


  let emailError = document.getElementById("emailError");
  let phoneError = document.getElementById("phoneError");


  const emailInput = document.getElementById("email");
  const phoneInput = document.getElementById("phone");


  let isValid = true;


  emailError.textContent = "";
  phoneError.textContent = "";
  emailInput.classList.remove("error-border", "success");
```

```
phoneInput.classList.remove("error-border", "success");

if (email === "") {
    emailError.textContent = "Email is required";
    emailInput.classList.add("error-border");
    isValid = false;
} else if (!validator.isEmail(email)) {
    emailError.textContent = "Enter a valid email";
    emailInput.classList.add("error-border");
    isValid = false;
} else {
    emailInput.classList.add("success");
}
if (phone === "") {
    phoneError.textContent = "Phone number is required";
    phoneInput.classList.add("error-border");
    isValid = false;
} else if (!validator.isMobilePhone(phone, 'en-IN')) {
    phoneError.textContent = "Enter a valid Indian phone number";
    phoneInput.classList.add("error-border");
    isValid = false;
} else {
    phoneInput.classList.add("success");
}

if (isValid) {
```

```
        alert("Form submitted successfully!");
    }
});
```

Output:





RESULT:

The input form was successfully designed and validated using HTML, CSS, and JavaScript. The system effectively checks user inputs and provides both textual and visual feedback, ensuring accurate data entry.

CONCLUSION:

This experiment demonstrates the importance of client-side validation in web applications. Using Validator.js simplifies validation logic and improves reliability. The addition of visual feedback enhances usability and provides a better user experience.